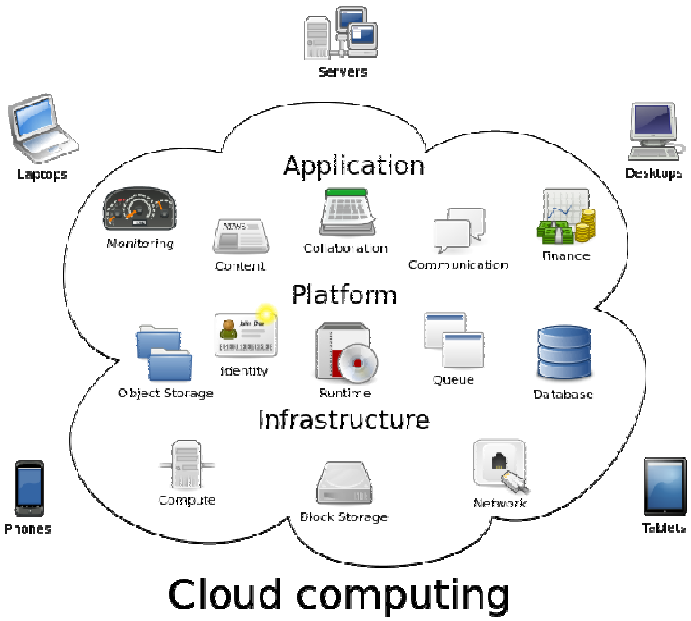
Defining the Cloud

The term cloud has been used as a metaphor for the Internet. This usage was originally derived from its common depiction in network diagrams as an outline of a cloud, used to represent the transport of data across carrier backbones (which owned the cloud) to an endpoint location on the other side of the cloud.



The cloud is a set of hardware, networks, storage, services, and interfaces that enable the delivery of computing as a service. Cloud services include the delivery of software, infrastructure, and storage over the internet (either as separate components or a complete platform) based on user demand.

Cloud computing is a pay-per-use model for enabling available, convenient, on-demand network access to a shared pool or configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

Key characteristics of Cloud Computing:

- **On-demand self-service**

  A consumer can independently provision computing capabilities such as server time and network storage, as needed without requiring human interaction with each service's provider.

- **Ubiquitous(easily available from everywhere) network access**

  Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms including mobiles, laptops etc.

- **Location-independent resource pooling**

  The consumer generally has no control over or knowledge of the exact location of the provided resources like storage, processing, memory, bandwidth etc.

- **Rapid elasticity**

  Capabilities can be rapidly and elastically provisioned to quickly scale up, and rapidly released to quickly scale down. To the consumer, the capabilities available for rent often appear to be infinite and can be purchased in any quantity at any time.

- **Pay-per-use**

  Capabilities are charged using a metered, fee-for-service, or advertising based billing model to promote optimization of resource use.

Basic building blocks of Cloud:

- **The end user**
  The end user doesn't really have to know anything about the underlying technology. In small businesses, for example, the cloud provider becomes the de facto data center. In larger organizations, the IT organization oversees the inner workings of both internal resources and external cloud resources.
- **Business management**
  Business management needs to take responsibility for overall governance of data or services living in a cloud. Cloud service providers must provide a predictable and guaranteed service level and security to all their constituents.
- **The cloud service provider**
  The cloud service provider is responsible for IT assets and maintenance.

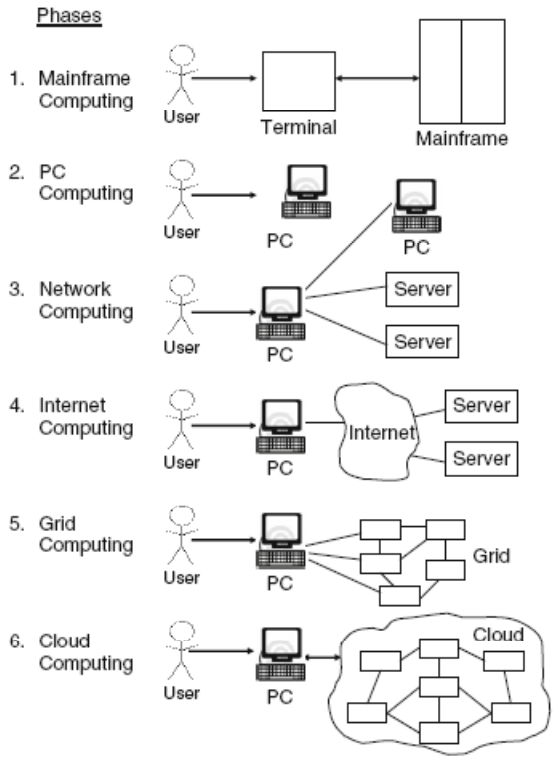
The Emergence of Cloud Computing

The underlying concept of cloud computing dates back to the 1950s; when large-scale mainframe became available in academia and corporations, accessible via thin clients /terminal computers.

In the 1990s, telecommunications companies, who previously offered primarily dedicated point-to-point data circuits, began offering virtual private network (VPN) services with comparable quality of service but at a much lower cost. By switching traffic to balance utilization as they saw fit, they were able to utilize their overall network bandwidth more effectively.

One of the first milestones for cloud computing was the arrival of Salesforce.com in 1999, which pioneered the concept of delivering enterprise applications via a simple website. The services firm paved the way for both specialist and mainstream software firms to deliver applications over the internet.

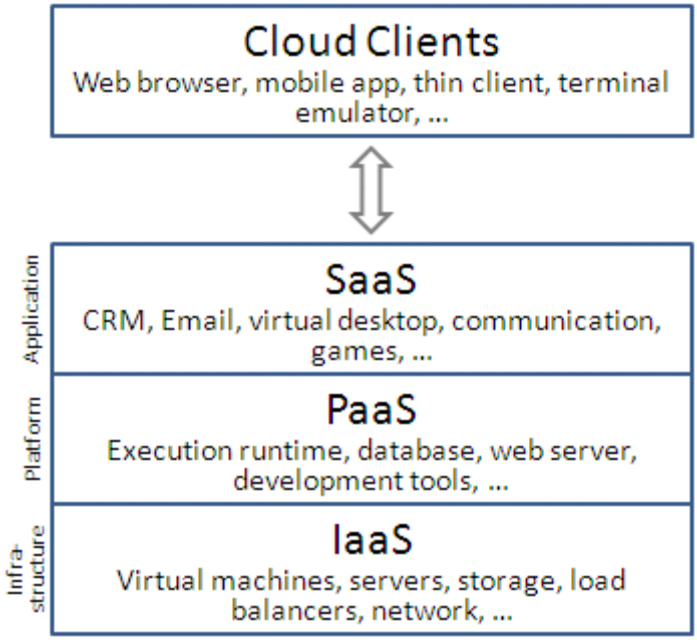
The next development was Amazon Web Services in 2002, which provided a suite of cloud-based services including storage, computation and even human intelligence through the Amazon Mechanical Turk. Then in 2006, Amazon launched its Elastic Compute cloud (EC2) as a commercial web service that allows small companies and individuals to rent computers on which to run their own computer applications.



Azure was announced in October 2008 and released on 1 February 2010 as Windows Azure, before being renamed to Microsoft Azure on 25 March 2014. It provides both PaaS and IaaS services and supports many different programming languages, tools and frameworks, including both Microsoft-specific and third-party software and systems.

Cloud-Based Services

The services given by the service provider to the customers or users through cloud computing technology are said cloud services. Service Provider's server gives both the hardware and software necessary and thus easy in management for both user and the cloud service provider. Cloud computing providers offer their services according to three fundamental models: Infrastructure as a service (IaaS), Platform as a service (PaaS), and Software as a service (SaaS) where IaaS is the most basic and each higher model abstracts from the details of the lower models.



a. Infrastructure as a service (IaaS)

IaaS refers not to a machine that does all the work, but simply to a facility given to businesses that offers users the leverage of extra storage space in servers and data centers. In this most basic cloud service model, providers offer computers, as physical or more often as virtual machines, and other resources. . Other resources in IaaS clouds include images in a virtual machine image library, raw (block) and file-based storage, firewalls, load balancers, IP addresses, virtual local area networks (VLANs), and software bundles. Examples of IaaS include: Amazon CloudFormation (and underlying services such as Amazon EC2), Rackspace Cloud, Terremark, Windows Azure Virtual Machines, Google Compute Engine, and Joyent.

b. Platform as a service (PaaS)

In the PaaS model, cloud providers deliver a computing platform typically including operating system, programming language execution environment, database, and web server. Application developers can develop and run their software solutions on a cloud platform without the cost and complexity of buying and managing the underlying hardware and software layers. Examples of PaaS include: Amazon Elastic Beanstalk, Cloud Foundry, Heroku, Force.com, EngineYard, Mendix, Google App Engine, Windows Azure Compute and OrangeScape.

c. Software as a service (SaaS)

In this model, cloud providers install and operate application software in the cloud and cloud users access the software from cloud clients. The cloud users do not manage the cloud infrastructure and platform on which the application is running. This eliminates the need to install and run the application on the cloud user's own computers simplifying maintenance and support. The pricing model for SaaS applications is typically a monthly or yearly flat fee per user, so price is scalable and adjustable if users are added or removed at any point. Examples of SaaS include: Google Apps, Microsoft Office 365, and Onlive.

Grid Computing or Cloud Computing?

Grid computing is often confused with cloud computing.

Grid computing is where more than one computer coordinates to solve a problem together. Often used for problems involving a lot of number crunching, which can be easily parallelisable.

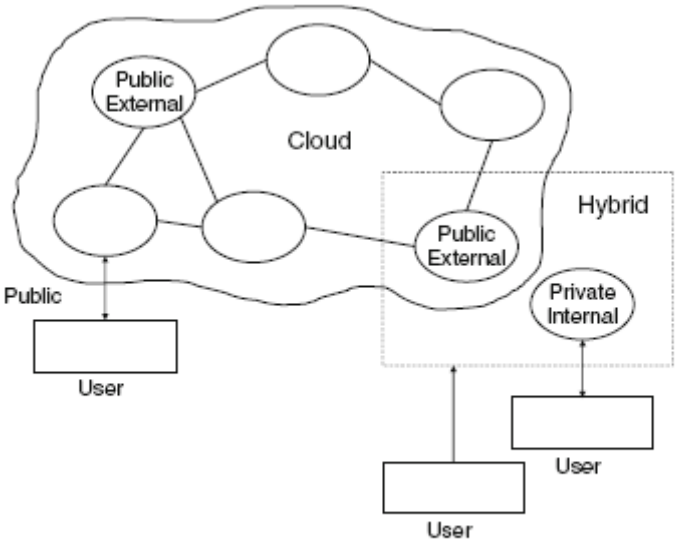
Cloud computing is where an application doesn't access resources it requires directly, rather it accesses them through something like a service. So instead of talking to a specific hard drive for storage, and a specific CPU for computation, etc. it talks to some service that provides these resources. The service then maps any requests for resources to its physical resources, in order to provide for the application. Usually the service has access to a large amount of physical resources, and can dynamically allocate them as they are needed.

A cloud would usually use a grid. A grid is not necessarily a cloud or part of a cloud.

Several deployment models for cloud computing

- **Private cloud**

  The infrastructure is operated solely for an organization; it may be managed by the organization or a third party and may exist on or off the premises of the organization. It is also referred to as 'corporate' or 'internal' Cloud, term used to denote a proprietary computing architecture providing hosted services on private networks.



- **Public cloud**

  The infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services. It is also referred to as 'external' Cloud that describes the conventional meaning of Cloud computing: scalable, dynamically provisioned, often virtualized resources available over the internet from an off-site third party provider, which divides up resources and bills its customers on a 'utility' basis.

- **Community cloud**

  Community cloud shares infrastructure between several organizations from a specific community with common concerns (security, compliance, jurisdiction, etc.), whether managed internally or by a third-party and hosted internally or externally. The costs are spread over fewer users than a public cloud (but more than a private cloud), so only some of the cost savings potential of cloud computing are realized.

- **Hybrid cloud**

  Here, the infrastructure is a composition of two or more clouds (private, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds). Cloud bursting is an application deployment model in which an application runs in a private cloud or data center and bursts into a public cloud when the demand for computing capacity spikes. The advantage of such a hybrid cloud deployment is that an organization only pays for extra compute resources when they are needed.

Benefits of using cloud models

- **Cost Savings**

  Companies can reduce their capital expenditures and use operational expenditures for increasing their computing capabilities. This is a lower barrier to entry and also requires fewer in-house IT resources to provide system support.

- **Scalability/Flexibility**

  Companies can start with a small deployment and grow to a large deployment fairly rapidly, and then scale back if necessary. Also, the flexibility of cloud computing allows companies to use extra resources at peak times, enabling them to satisfy consumer demands.

- **Reliability**

  Services using multiple redundant sites can support business continuity and disaster recovery.

- **Maintenance**

  Cloud service providers do the system maintenance, and access is through APIs that do not require application installations onto PCs, thus further reducing maintenance requirements.

- **Mobile Accessible**

  Mobile workers have increased productivity due to systems accessible in an infrastructure available from anywhere.

Characteristics of Cloud Computing

Cloud computing has a variety of characteristics, with the main ones being:

- **Shared Infrastructure**

  Uses a virtualized software model, enabling the sharing of physical services, storage, and networking capabilities. The cloud infrastructure, regardless of deployment model, seeks to make the most of the available infrastructure across a number of users.

- **Dynamic Provisioning / on demand self service**

  Allows for the provision of services based on current demand requirements. This is done automatically using software automation, enabling the expansion and contraction of service capability, as needed. This dynamic scaling needs to be done while maintaining high levels of reliability and security.

- **Broad Network Access**

  Needs to be accessed across the internet from a broad range of devices such as PCs, laptops, and mobile devices, using standards-based APIs (for example, ones based on HTTP). Deployments of services in the cloud include everything from using business applications to the latest application on the newest smartphones.

- **Multi-Tenant Capable**

  The resources (e.g., network, storage and compute power) can be shared among multiple enterprise clients, thereby lowering overall expense. Resource virtualization is used to enforce isolation and aid in security.

- **Rapid Elasticity**

  The consumer should have the ability to rapidly (often automatically) increase or decrease the computing resources needed to carry out their work.

- **Managed Metering / Measured Service**

  Uses metering for managing and optimizing the service and to provide reporting and billing information. In this way, consumers are billed for services according to how much they have actually used during the billing period.

Challenges for cloud computing

- **Security and Privacy**

  These issues are generally attributed to slowing the deployment of cloud services. These challenges can be addressed, for example, by storing the information internal to the organization, but allowing it to be used in the cloud. For this to occur, though, the security mechanisms between organization and the cloud need to be robust and a Hybrid cloud could support such a deployment.
- **Lack of Standards**

  Clouds have documented interfaces; however, no standards are associated with these, and thus it is unlikely that most clouds will be interoperable.
- **Continuously Evolving**

  User requirements are continuously evolving, as are the requirements for interfaces, networking, and storage. This means that a "cloud," especially a public one, does not remain static and is also continuously evolving.
- **Compliance Concerns**

  The EU has a legislative backing for data protection across all member states, but in the US data protection is different and can vary from state to state. As with security and privacy mentioned previously, these typically result in Hybrid cloud deployment with one cloud storing the data internal to the organization.

Grid Computing

Grid computing is a processor architecture that combines computer resources from various domains to reach a main objective. In grid computing, the computers on the network can work on a task together, thus functioning as a supercomputer.

Typically, a grid works on various tasks within a network, but it is also capable of working on specialized applications. It is designed to solve problems that are too big for a supercomputer while maintaining the flexibility to process numerous smaller problems. Computing grids deliver a multiuser infrastructure that accommodates the discontinuous demands of large information processing.

The idea of grid computing was first established in the early 1990s by Carl Kesselman, Ian Foster and Steve Tuecke. They developed the Globus Toolkit standard, which included grids for data storage management, data processing and intensive computation management.

Grid computing is made up of applications used for computational computer problems that are connected in a parallel networking environment. It connects each PC and combines information to form one application that is computation-intensive.

Distributed Computing in Grid and Cloud

Distributed computing is a field of computer science that studies distributed systems. A distributed system consists of multiple autonomous computers that communicate through a computer network. The computers interact with each other in

order to achieve a common goal. In distributed computing, a problem is divided into many tasks, each of which is solved by one or more computers, which communicate with each other by message passing. While there is no single definition of a distributed system, the following defining properties are commonly used:

- There are several autonomous computational entities, each of which has its own local memory.
- The entities communicate with each other by message passing

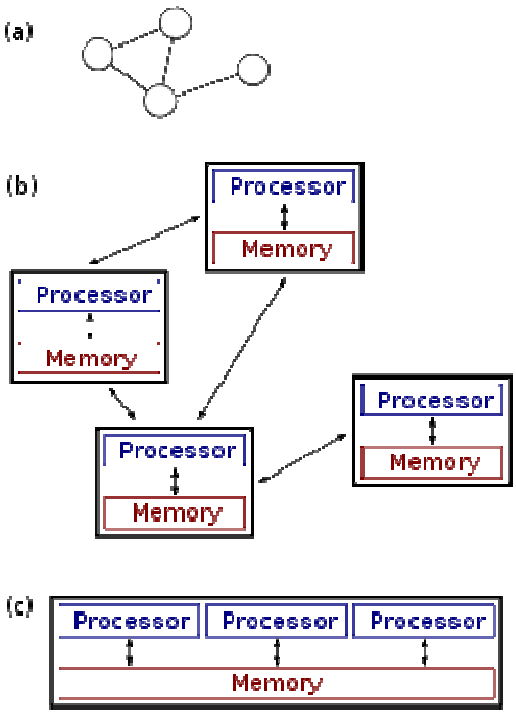



Figure (a) is a schematic view of a typical distributed system; as usual, the system is represented as a network topology in which each node is a computer and each line connecting the nodes is a communication link. Figure (b) shows the same distributed system in more detail: each computer has its own local memory, and information can be exchanged only by passing messages from one node to another by using the available communication links. Figure (c) shows a parallel system in which each processor has a direct access to a shared memory.